

Breakthrough therapy designation of nirsevimab for the prevention of lower respiratory tract illness caused by respiratory syncytial virus infections (RSV).

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Respiratory syncytial virus (RSV) is a leading cause of serious lower respiratory tract infection (LRTI) in infants and young children. Palivizumab is an RSV-specific prophylactic for use in high-risk infants but treatment requires monthly injections and only modestly reduces hospitalization. Thus, new immunoprophylactic candidates are under development. Nirsevimab (MEDI8897) is a monoclonal antibody with an extended half-life developed to protect infants for an entire RSV season with a single dose. This review summarizes clinical trial data on nirsevimab. The authors introduce RSV and surface viral proteins involved in infection, then discuss the development and achievements of nirsevimab in clinical trials concluding with expert opinion. Information was compiled from PubMed, clinicaltrials.gov, and press releases from AstraZeneca and Sanofi. Nirsevimab (MEDI8897) is an RSV F protein monoclonal antibody and the next-generation RSV medicine having an extended half-life developed for the prevention of LRTI caused by RSV. Nirsevimab will supplant the current standard of care for RSV prevention. Importantly, nirsevimab requires a single dose to last the entire RSV season and may be given to term, preterm, and high-risk infants. However, even with nirsevimab approval there remains a need for an efficacious RSV vaccine and treatments.

Parent Perspectives on Nirsevimab for Their Newborn.

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In 2023, nirsevimab, a monoclonal antibody against respiratory syncytial virus (RSV), was approved in the United States to be given to infants to prevent bronchiolitis, a leading cause of pediatric hospitalizations. We sought to better understand how parents of otherwise healthy newborns would respond to this recommendation. We conducted semistructured interviews of purposefully sampled parents of newborns admitted to the well newborn unit between November 2023 and February 2024. Interviews were performed in a constant comparative manner until thematic sufficiency was reached using the Health Beliefs Model as a conceptual framework. We conducted 28 interviews: 38% of participants planned to give nirsevimab to their newborn, 25% did not plan to, and 38% were unsure. Four major themes emerged: (1) Major knowledge gaps existed about RSV prophylaxis; (2) trust in pediatrician's recommendation and fear of RSV infection contributed to uptake; (3) parents deferring RSV prophylaxis were concerned about side effects, wanted more time to decide, felt the maternal RSV vaccine was sufficient, and trusted their own prevention measures; and (4) there was misinformation about nirsevimab among parents including a perception that nirsevimab is a new understudied vaccine with similarities to the COVID-19 vaccine. Application of the Health Beliefs Model highlighted actionable opportunities to positively influence decision making around perceived susceptibility and RSV disease severity, benefits, barriers, self-efficacy, and cues to action. Increasing overall awareness of both nirsevimab and RSV in infants, starting conversations during pregnancy to combat misinformation, leveraging trusting relationships with pediatricians and prenatal clinicians can help with future uptake.

Extracranial metastasis of brain glioblastoma outside CNS: Pathogenesis revisited.

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The most prevalent malignant tumor of the CNS in adults is glioblastoma. Despite undergoing surgery and chemoradiotherapy, the prognosis remains unfavorable, with a median survival period ranging between 15 and 20 months. The incidence of glioblastoma metastasis outside CNS is uncommon with only 0.4%-2% reported rate, compared to other tumors that exhibit a 10% incidence rate of metastasis to the brain. On average, it takes about 11 months from the time of initial diagnosis for the tumor to spread beyond CNS. Consequently, the prognosis for metastatic glioblastoma is grim, with a 6-month survival rate following diagnosis. The rarity of extracranial metastasis is attributed to the blood-brain barrier and lack of a lymphatic drainage system, although rare cases of hematogenous spread and direct implantation have been reported. The possible mechanisms remain unclear and require further investigation. Risk factors have been widely described, including previous craniotomy or biopsies, ventricular shunting, young age, radiation therapy, prolonged survival time, and tumor recurrence. Due to the lack of understanding about extracranial metastasis of glioblastoma pathogenesis, no effective treatment exists to date. Aggressive chemotherapies are not recommended for metastatic glioblastoma as their side effects may worsen the patient prognosis. The optimal treatment for extracranial metastasis of glioblastoma requires further investigation with a wide inclusion of patients. This review discusses the possible causes, factors, and underlying mechanisms of glioblastoma metastasis to different organs.

Why are systemic glioblastoma metastases rare? Systemic and cerebral growth of mouse glioblastoma.

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Systemic metastasis of glioblastoma multiforme (GBM) in the form of bulk tumor is rare. This could be because of patient death before clinically detectable systemic metastasis, impediments to systemic egress, or the inability of GBM to grow outside the central nervous system (CNS). In the present paper, we tested this last hypothesis. The delayed brain tumor (DBT) cell was characterized with respect to in vitro and in vivo morphology, growth rate, anchorage-independent growth, glial fibrillary acidic protein expression and cytogenetic analysis, and major histocompatibility complex (MHC) typing. We then assayed implantation-induced intracerebral and systemic GBM growth using 3 rodent models with increasing relative immunologic differences between implanted DBT cells and hosts (Balb/c mice, an isograft, MHC I H2, class type D; C3H mice, an allograft; Wistar rats, a xenograft). After implantation in the brain, DBT cells generated tumors that were similar to human GBM. Intracerebral DBT implantation as an isograft or allograft produced only intracranial tumors, whereas intracerebral and systemic implantation as a xenograft produced no tumors. Systemic isograft implantation yielded only systemic tumors. Systemic implantation as allografts produced only transient subcutaneous masses. Delayed brain tumor cells implanted outside the CNS formed tumors unless there was a significant difference between the immunotype of the implanted cells and host. These results support the hypothesis that the rarity of systemic GBM tumors lies in the presence of physical barriers and/or systemic hurdles that prevent their timely growth. These results also demonstrate that GBMs are antigenic, although not immunogenic, with their syngeneic host. Therefore, GBM may be amenable to targeted immunotherapy given successful artificial priming of the immune system.

Mesenchymal-Type Genetic Mutations Are Likely Prerequisite for Glioblastoma Multiforme to Metastasize Outside the Central Nervous System: An Original Case Series

and Systematic Review of the Literature.

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Glioblastoma multiforme (GBM) is the most aggressive and prevalent type of malignant brain tumor, yet it metastasizes outside the central nervous system (CNS) in only 0.4% of cases. Little is known about what enables this subset of GBMs to take root outside the CNS, but genetic mutations likely play a role. We conducted a PRISMA-compliant systematic review of metastatic GBM wherein we reviewed 3579 search results and 1080 abstracts, analyzing data from 139 studies and 211 unique patients. In addition, we describe 4 cases of patients with pathologically confirmed GBM metastases outside the CNS treated at our institution. We found that metastases were discovered near previous surgical sites in at least 36.9% of cases. Other sites of metastasis included bone (47.9%), lung (25.6%), lymph nodes (25.1%), scalp (19.2%), and liver (14.2%). On average, metastases were diagnosed 12.1 months after the most recent resection, and the mean survival from discovery was 5.7 months. In our patients, primary GBM lesions showed mutations in NF1, TERT, TP53, CDK4, and RB1/PTEN genes. Unique to the metastatic lesions were amplifications in genes such as p53 and PDGFRA/KIT, as well as increased vimentin and Ki-67 expression. There is strong evidence that GBMs acquire novel mutations to survive outside the CNS. In some cases, tumor cells likely mutate after seeding scalp tissue during surgery, and in others, they mutate and spread without surgery. Future studies and genetic profiling of primary and metastatic lesions may help uncover the mechanisms of spread.

Systemic dissemination of glioblastoma: literature review.

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Glioblastoma (GBM) is the most frequent primary malignant tumor from the central nervous system in adults. However, the presence of systemic metastasis is an extremely rare event. The objective of this study was to review the literature, evaluating the possible biological mechanisms related to the occurrence of systemic metastasis in patients diagnosed with GBM. The mechanisms that may be related to GBM systemic dissemination are the blood-brain barrier breach, often seen in GBM cases, by the tumor itself or by surgical procedures, gaining access to blood and lymphatic vessels, associated with the acquisition of mesenchymal features of invasiveness, resistance to the immune mechanisms of defense and hostile environment through quiescence. Tumor cells must overcome many obstacles until the development of systemic metastasis. The physiologic mechanisms are not completely clear. Although not fully understood, the pathophysiological understanding of the mechanisms that may be associated with the systemic spread is salutary for a global understanding of the disease. In addition, this knowledge may be used as a basis for a therapy to be performed in patients diagnosed with GBM distant metastasis.

Extraneural Metastases of Glioblastoma without Simultaneous Central Nervous System Recurrence.

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Glioblastoma multiforme (GBM) is well known as the most common malignant primary brain tumor. It could easily spread into the adjacent or distant brain tissue by infiltration, direct extension and cerebro-spinal fluid dissemination. The extraneural metastatic spread of GBM is relatively rare but it

could have more progressive disease course. We report a 39-year-old man who had multiple bone metastases and malignant pleural effusion of the GBM without primary site recurrence.

Diffuse bony metastases at presentation in a child with glioblastoma multiforme. A case report.

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At initial diagnosis, an 11-year-old girl with glioblastoma multiforme (GBM) presented with diffuse osteoblastic metastases. Primary brain tumors rarely metastasize outside of the central nervous system (CNS) without prior neurosurgery. Extracranial spread at diagnosis has been previously documented in just two adults. Extracranial metastasis of a childhood glioma without prior neurosurgery at any time during the course of the disease is exceedingly rare. Spread to bone by gliomas is also infrequent, and when they occur, bony metastases are usually isolated to one or two sites in any given patient. The widespread osseous metastases in our patient have been reported in three prior cases of high-grade gliomas. This child's GBM likely reflects a highly aggressive variant with the potential to spread outside the CNS and with a predilection for bone. Oncologists should be aware that GBM may present in this fashion during childhood.

Extraneural metastasis of glioblastoma multiforme presenting as an unusual neck mass.

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Glioblastoma multiforme (GBM) is the most aggressive intracranial tumor and it commonly spreads by direct extension and infiltration into the adjacent brain tissue and along the white matter tract. The metastatic spread of GBM outside of the central nervous system (CNS) is rare. The possible mechanisms of extraneural metastasis of the GBM have been suggested. They include the lymphatic spread, the venous invasion and the direct invasion through dura and bone. We experienced a 46-year-old man who had extraneural metastasis of the GBM on his left neck. The patient was treated with surgery for 5 times, radiotherapy and chemotherapy. He had survived 6 years since first diagnosed. Although the exact mechanism of the extraneural metastasis is not well understood, this present case shows the possibility of extraneural metastasis of the GBM, especially in patients with long survival.

A case series of osseous metastases in patients with glioblastoma.

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Extracranial metastases occur in <2% of cases of glioblastoma (GBM). When metastases do occur, bone is the most common destination. Herein, we review clinical characteristics of GBM patients with osseous metastases and evaluate both potential risk factors and prognostic significance. Using an institutional database, we identified and retrospectively analyzed 6 patients with both GBM and osseous metastases. We collected data on patient demographics, tumor genetics, clinical courses, and outcomes. Given the rarity of metastatic GBM, we conducted historical comparisons using previously published literature. Five patients with osseous metastases (83%) were male, with a median age of 46 years at GBM diagnosis

(range: 20-84). All patients had IDH-wildtype, MGMT promoter unmethylated GBM and 5 (83%) had alterations in TP53. All patients underwent surgical resection for GBM followed by radiation with concurrent and adjuvant temozolomide. Four patients (67%) received bevacizumab prior to bone metastasis diagnosis. Bone metastases were discovered at a median of 12.2 months (range: 5.3-35.2) after GBM diagnosis and 4.8 months after starting bevacizumab (range: 3.5-13.2). Three patients (50%) received immunotherapy. After osseous metastasis diagnosis, the median survival was 25 days (range: 13-225). In our cohort, most patients were male and young at the time of GBM diagnosis. All patients had IDH-wildtype, MGMT promoter unmethylated GBM, and most had alterations in TP53, which may be important for osseous metastasis. Most patients received bevacizumab, which has been associated with earlier metastasis. Osseous metastases of GBM occur and portend a dismal prognosis in an already aggressive malignancy.
